0765/3/2025
P.M.M. A/L

SOUTH WEST REGIONAL MOCK EXAMINATION
GENERAL EDUCATION

THE TEACHERS' RESOURCE UNIT (TRU)

IN COLLABORATION WITH

THE REGIONAL INSPECTORATE OF PEDAGOGY FOR SCIENCE

AND

THE SOUTH WEST ASSOCIATION OF MATHEMATICS TEACHERS (SWAMT)

Saturday 29/03/2025 - Afternoon ADVANCED LEVEL

Subject Title	**PURE MATHEMATICS WITH MECHANICS**
Subject Code Number	**0765**
Paper Number	**Paper 3**

THREE HOURS

INSTRUCTIONS TO CANDIDATES

Answer ALL questions.

For your guidance, the approximate mark allocation for parts of each question is indicated in brackets.

You are reminded of the necessity for good English and orderly presentation in your answers.

Mathematical formulae and tables published by the GCE Board and noiseless, non-programmable calculators are allowed.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

Where necessary, take g as $10\ m\ s^{-2}$.

1. The position vector of a particle P of mass 3 kg at time t seconds is

$$\mathbf{r} = [(3 + \sin 2t)\mathbf{i} + (\cos 2t)\mathbf{j}] \text{ m}$$

(a) Find the value of t when P crosses the x-axis for the first time. **(3 marks)**

(b) Find the velocity and acceleration of P at time t. **(4 marks)**

(c) Show that the speed of the particle at any time is constant. **(3 marks)**

(d) Find the magnitude of the force acting on the particle when $t = \frac{\pi}{2}$ **(3 marks)**

---

2. (i) A system is made up of the forces

$$\mathbf{F}_1 = (3\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) \text{ N}, \quad \mathbf{F}_2 = (2\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}) \text{ N} \quad \text{and} \quad \mathbf{F}_3 = (-5\mathbf{i} + \mathbf{k}) \text{ N}$$

acting at the points with position vectors

$$\mathbf{r}_1 = (\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) \text{ m}, \quad \mathbf{r}_2 = (3\mathbf{i} - \mathbf{k}) \text{ m} \quad \text{and} \quad \mathbf{r}_3 = (6\mathbf{i} + 2\mathbf{j} + \mathbf{k}) \text{ m respectively.}$$

Show that the system is in equilibrium. **(5 marks)**

(ii) A non uniform ladder AB of length $8a$ and weight W is such that its weight acts at the point G on the ladder, where $AG = \frac{7a}{2}$. The ladder rests with the end A on rough horizontal ground and the end B on smooth vertical wall, making an angle of 45^o with the horizontal. The coefficient of friction between the ladder and the ground is $\frac{1}{2}$. A man of weight $4W$ stands at a point on the ladder, a distance $5a$ from the foot. His son, of weight $2W$ also starts climbing slowly. Show that the ladder begins to slip when the distance on the ladder between the man and his son is $\frac{11a}{4}$ **(8 marks)**

---

3. (i) A smooth sphere of mass 2 kg is moving with velocity $(2\mathbf{i} - 3\mathbf{j})$ ms^{-1} when it is struck by a smooth surface. As a result of this impact, the sphere receives an impulse of $(4\mathbf{i} + \lambda\mathbf{j})$ kgms^{-1} and moves with velocity $(\mu\mathbf{i} + 8\mathbf{j})$ ms^{-1}. Find the values of the constants λ and μ. **(4 marks)**

(ii) A smooth sphere A of mass m moving on a smooth horizontal table with velocity $2u$ when it collides directly with another smooth sphere B of mass $3m$ moving in the same direction as A on the same table with velocity u. After the collision, B moves with velocity ku, in the same direction, where k is a constant.

Find, in terms of u and k,

(a) the velocity of A immediately after the collision. **(3 marks)**

(b) the coefficient of restitution between the spheres. **(3 marks)**

(c) Hence, or otherwise, deduce that $\frac{5}{4} \leq k \leq \frac{3}{2}$ **(3 marks)**

---

4. The engine of a motorbike of mass 300 kg is working at a constant rate of 54kW, against a total non-gravitational resistance of magnitude $(a + bv)$ N, where v is its speed, in metres per second and a and b are constants. The maximum speed of the motorbike on a horizontal stretch of road is 60 ms^{-1} while its maximum speed on the stretch of road inclined at $\sin^{-1}\left(\frac{2}{25}\right)$ to the horizontal is 50 ms^{-1}.

(a) Find the values of the constants a and b. **(9 marks)**

(b) Determine the acceleration of the motorbike on a straight level road at the instant when its speed is 30 ms^{-1}. **(4 marks)**

---

5. A particle of mass 0.5 kg is projected with speed u from a point, O, on level horizontal ground, 40 m away from the foot of a vertical wall. The particle just clears the wall and another parallel wall before striking the ground again at the point A. The distance between the walls is 40 m and each wall has a height of 20 m. The plane in which the particle travels is perpendicular to the walls.

(a) Find the angle of projection of the particle. **(5 marks)**

(b) Show that $u = 25\sqrt{2}$. **(3 marks)**

(c) Calculate the kinetic energy of the particle just when it reaches maximum height. **(5 marks)**

---

for more past questions and solutions download kawlo kawlo app or visit http://www.gcerevision.com

6. (i) A particle A of mass 2 kg rests on a smooth plane inclined at 30^o to the horizontal. It is connected by a light inextensible string passing over a smooth pulley fixed at the top of the plane, to another particle B of mass m. The system is released from rest with both parts of the string taut and the hanging part vertical. Given that A moves up the plane with an acceleration of $\frac{5}{2}$ m s^{-2}, find the:

(a) Tension in the string. **(5 marks)**

(b) Value of m. **(2 marks)**

(c) Magnitude of the reaction force at the pulley. **(2 marks)**

(ii) A particle of mass m moves with constant speed u in a horizontal circle of radius a on the inner surface of a smooth spherical bowl of internal radius $2a$.

Show that $u^2\sqrt{3} = ag$. **(4 marks)**

7. (i) Show, by integration, that the centroid of the uniform lamina bounded by the curve $y = 4 - x^2$ from $x = 0$ to $x = 2$ is at the point $\left(\frac{3}{4}, \frac{8}{5}\right)$. **(5 marks)**

(ii) In Figure 1 below, $OABC$ is a rectangular lamina. The arc CD of the curve $y = 4 - x^2$ is cut off from the lamina $OABC$ as shown.

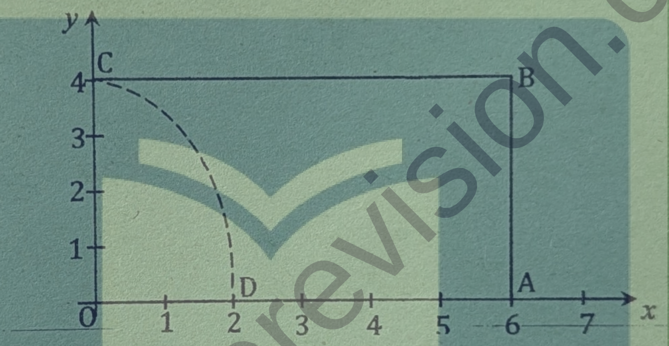

Figure 1

(a) Find the coordinates of the centroid of the remainder $ABCD$. **(6 marks)**

The remaining lamina $ABCD$ is suspended freely from B.

(b) Find the angle of inclination of AB to the vertical. **(2 marks)**

8. (i) Four letters are chosen at random from the word **READING**. Find the probability of choosing

(a) Exactly two vowels **(2 marks)**

(b) At least two vowels. **(3 marks)**

(ii) A bag contains 3 red balls and 5 green balls. A ball is drawn at random from the bag, its color noted and replaced together with two other balls of the other color. Another ball is then drawn at random from the bag.

(a) Draw a probability tree diagram showing all the outcomes and probabilities. **(3 marks)**

Hence, or otherwise, find the probability of drawing

(b) a red and a green ball in that order. **(2 marks)**

(c) two balls of the same color. **(3 marks)**

END